A. Supplementary tables

Table S1: Representative data-level fusion studies. Validation evidence is defined in Section 6.4.

Paper	Year	Modalities	Method	Datasets	Metrics	Validation evidence
Fusion system[49]	1993	CT-MR	Color channel mapping	Private	Point displacement error, spatial dimension accuracy, visual evaluation	Unclear
3D voxel fusion[50]	2004	CT-MRI, MRI multi-sequences	3D reconstruction, color coding	Private	Fusion accuracy, FPS	Unclear
3D spinal fusion[51]	2004	3D CT-MRI	Registration, voxel merging, Semi-Boundary	Private	Visual evaluation	Unclear
Color table fusion[149]	2004	CT-PET-MRI	Anatomy specific color fusion, Observer oriented fusion	Private	Visual evaluation	Unclear
Regis+ICP[150]	2007	DE-MRI&PV-MRA, PV-MRA&EAM, DE-MRI&PV-MRA&EAM	Intensity-based rigid registration, ICP	Private	Registration error	Unclear
TRUS-MRI fusion[52]	2007	TRUS-MRI	Manual registration method for 3D US reconstruction	Private	Hausdorff, TRE, FRE	Unclear
AWT+curvelet transform[151]	2010	MRI-CT	Curvelet transform	Private	Similarity, PSNR	Unclear
PCA+DWT[152]	2013	CT-PET	PCA, DWT	Private	MSE, PSNR, Entropy, visual evaluation	Unclear
Improved ICP[153]	2014	CT& Blue-Ray	ICP, improved Voronoi	Private	Registration average Euclidean distance	Unclear
DCxWT [154]	2014	CT-MRI, T1w MRI-MRA	DCxWT, multi-scale max fusion	**Imagefusion.org**	Entropy, Q^F_{AB}, FF, SD, FS	Unclear
Block PCNN[155]	2015	CT-MRI	S-PCNN, OFG, Image block processing	Private	FD, MI, Q^F_{AB}, MV, RMSE	Unclear
QC hybrid[156]	2015	CT-MRI	Pseudo conformal mapping, mixed criteria	Private	e_{max}/e_{mean}, Beltrami, MI, visual evaluation	Unclear
CMA-ES[157]	2016	CT-MRI	Morphological filtering, evolutionary optimization	Private	Registration error, time, visual evaluation	Unclear
Union LAP[158]	2016	CT-MRI, PET-MRI, SPECT-MRI	Laplacian pyramid multi-scale decomposition, multi feature fusion, image statistics	**AANLIB**	PSNR, STD, SSIM, TMQI	Benchmark
QPSO-PCNN [159]	2016	CT-MR, MR T1-FDG, MR T2-SPET	QPSO-PCNN	Private	STD, SF, MI, SSIM, Sharpness	Unclear
Wavelet channel fusion[160]	2017	MR T1-CT, MR DWI-CT	Wavelet-Pseudo color-α channel	Human Connectome Project	IE, MI, MG, SF, visual evaluation	Unclear
MRR[123]	2017	CT-MRI	B-spline deformable registration	Kasturba Hospital	MI, visual evaluation	Reader study (radiologist ratings)
PCA-DWT[161]	2017	CT-MRI, MRI-SPECT	PCA, DWT, PCNN	**AANLIB**	PSNR, STD, Entropy, MI, Q^F_{AB}, visual evaluation	Unclear
LR sparse dictionary fusion[162]	2018	CT-MRI, MRI-T1/MRI-T2, MRI-PET, SPECT-MRI	Discriminative low-rank sparse dictionary learning, low-rank and sparse component decomposition, weighted nuclear norm regularization, max-absolute component fusion	**AANLIB**, **imagefusion.org**, private training samples	MI, Q_{TE}, Q_{NCIE}, Q_M, Q_P, Q_S, Q_G	Unclear
CNN Type-I (input)[32]	2018	PET-CT-MRI T1/T2	2D patch stacking across modalities; channel-spanning kernels	**TCIA**	Pixel-wise Accuracy, visual evaluation	Benchmark
Fuzzy logic[54]	2018	CT-MRI-PD	T-S fuzzy reasoning system	**AANLIB**	MSE, PSNR, SNR, visual evaluation	Unclear
Tetrolet SUnSAL [63]	2018	CT-MRI, PET-MRI, SPECT-MRI	Tetrolet transform, SUnSAL, cycle-spinning, MDC, WSML	**Imagefusion.org**, **AANLIB**	Visual evaluation, LSSIM, MI, Q^F_{AB}, FSIM/FSIMC	Unclear
NSST PCNN[163]	2018	CT-MRI T2	NSST, PCNN	PGIMER, **AANLIB**	STD, MI, SF, Q^F_{AB}, Mean	Benchmark; Unclear (cohort design)
NSST S-PCNN[164]	2018	CT-MRI-PET	HSV, NSST, ICM S-PCNN	**AANLIB**	AG, SF, MV, SD	Unclear
DRT[165]	2018	MRI-PET	DRT, SSF	**AANLIB**	MI, SF, ASSIM, Q^F_{AB}	Benchmark

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Table S1 continued

Paper	Year	Modalities	Method	Datasets	Metrics	Validation evidence
S-ICP [166]	2019	3D coronary artery-vein anatomy SPECT	S-ICP, CRT	Private	Registration error	Unclear
BWT [167]	2019	CT-MRI, MRI T1-MRA	BWT, GFS rules	**Imagefusion.org**	FF, FS, Entropy, Q^F_{AB}	Unclear
CS-MCA[168]	2019	CT-MRI T2, MRI T1-T2	CSR, MCA	**AANLIB**	LMI, Q^F_{AB}, Q_{HVS}, Q_E	Unclear
CNN-DCSCM[169]	2019	CT-MRI	VGG, DCSCM	**AANLIB**	MSSIM, IE, MI, SF, Q^F_{AB}	Benchmark
Siamese CNN[170]	2020	MRI-CT, MRI T1-T2, MRI-PET/SPECT	CNN, contrast pyramid	**AANLIB**	TE, MI, VIF, Q^F_{AB}, visual evaluation	Unclear
PMGI [171]	2020	MRI-PET	PGMI	**AANLIB**	CC, EN, FMI, MI, SF, SSIM, SD	Benchmark
MIEE-PCNN[172]	2021	CT-MRI, PET-MRI	MIEE, PCNN, PSO	**AANLIB**, **Imagefusion.org**	Visual evaluation, NMI, SSIM, $Q^{AB/F}$, FSIM	Unclear
BMGAN [56]	2021	MRI-PET	3D Dense-UNet, GAN	**ADNI**, OASIS-3, **TCIA**	PSNR, FID, MS-SSIM, MAE	Unclear
AISA-NSST [173]	2021	CT-PET, MRI-CT, MR T1-T2	NSST, AISA, MG	**AANLIB**	PSNR, SSIM, SD, EN, MI, $Q^{AB/F}$, time, visual evaluation	Benchmark
ECLAHE [174]	2021	CT-MRI	SNNE, ECLAHE	**AANLIB**, **BraTS 2015**	MSE, PSNR, SSIM, EN, F1, accuracy, recall	Benchmark
CS-PCA[175]	2021	MRI DW-FLAIR, DW-T1, FLAIR-T1, DW-FLAIR-T1	CS, PCA, Tsallis WSS	ISLES 2015	DSC, JI, PR, AC, SE	Benchmark
DCT [53]	2021	CT-MRI	DWT, DCT	Private	MSE, AG, PSNR, EN, SD, SNR	Unclear
Lambert[176]	2022	MRI-Fundus	2D-3D Lambert	Private	NCC, VR, AR, distance, visual evaluation	Unclear
Variational optimization[177]	2022	CT-MRI	Variational optimization fusion	Harvard Whole Brain Atlas	MI, Q_W, Q^G	Benchmark
IST-NSST [178]	2022	PET-CT	WAGLE, IST	**AANLIB**	VIF, EDGE, FD, AG, SF, SD	Unclear
CDF 9/7 NSST[179]	2022	CT-MRI, MRI-PET, MRI-SPECT	CDF 9/7 NSST, MPCNN	**AANLIB**	VIF, SS, EN, SD	Unclear
SIFI [103]	2022	CT-MRI, MRI T1-T2, MRI-Gad-T2	SIFS, fuzzy set	**AANLIB**	AG, MI, Entropy, SD, Mean, visual evaluation	Unclear
CBF-DTF-RGF[58]	2022	CT-MRI	DTF, RGF, CBF	MATLAB Central File Exchange	CC, EN, SD, MI, AG, API	Benchmark
SwinFusion [34]	2022	PET-MRI, CT-MRI	Swin Transformer, CNN	**AANLIB**	PSNR, SSIM, Q^F_{AB}, FMI	Benchmark
PAULPCNN [180]	2023	PET-MRI, SPECT-MRI	PAULPCNN, NSST, DWRE	**AANLIB**	UQI, Edge, AG, MIF, SSIM, Q^F_{AB}	Unclear
M3AE [12]	2023	MRI T1-T1c-T2-FLAIR	M^3AE, model inversion, self-distillation	**BraTS 2018, 2020**	DSC, HD95	Benchmark
3D U-Net [55]	2024	MRI T1-T1ce-T2-FLAIR	3D U-Net, attention	**BraTS 2018**	DSC, IoU	Unclear
MHFN input baseline [107]	2024	MRI T1W–T2W-STIR	Raw-input concatenation; baseline in the MHFN study	Private	DSC, IoU, HD	Unclear (single-center cohort)

Table S2: Representative feature-level fusion studies. Validation evidence is defined in Section 6.4.

Paper	Year	Modalities	Method	Datasets	Metrics	Validation evidence
Shunt color[181]	2001	MRI-PD, MRI-SPECT	Shunt Neural Network, Color Opponent Fusion	**AANLIB**	Visual evaluation	Unclear
Wavelet transform [182]	2001	PET-MRI, CT-MRI	Wavelet Pyramid, Mallat Orthogonal Wavelet Decomposition, Max Selection	Private	Bias, visual evaluation	Unclear
3D convolution[183]	2002	MRI PD-T1-T2, MRI-SPECT	Shunt Neural Network, Color Opponent Fusion	**AANLIB**	Visual evaluation	Unclear
Color table fusion [149]	2004	MRI-CT-PET	2D CLT, blending	Private	Visual evaluation	Unclear

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Table S2 continued

Paper	Year	Modalities	Method	Datasets	Metrics	Validation evidence
PCNN [184]	2005	CT-MRI, CT-SPECT, MRI-PET	DWF, PCNN, OMP	**AANLIB**	MI, EN, Visual evaluation	Unclear
2d DWT[59]	2006	CT-MRI	DWT, IDWT	Private	RMSE, PSNR	Unclear
Regis+ICP[150]	2007	DE-MRI&PV-MRA, PV-MRA&EAM, DE-MRI&PV-MRA&EAM	ICP	Private	Registration error	Unclear
Wavelet transform[185]	2010	CT-MRI, MRI-PET	DWT, Max Selection	Private	Mean, Variance, EN, CEN, visual evaluation	Unclear
FDCT [186]	2012	CT-MRI	DWT, FDCT, IDWT, IHS Transform	Private	RMSE, PSNR	Unclear
NSCT [187]	2013	CT-MRI	NSCT, MSML	Private	MI, $Q^{AB/F}$, SF	Unclear
DWT-PCA [152]	2013	CT-PET	PCA, DWT	Private	MSE, PSNR, EM, visual evaluation	Unclear
Framelet HVS[188]	2013	CT-MRI, MRI-PET	Framelet Transform, HVS, IHS Transform	**AANLIB**	Mean, EN, SF, MI, SSIM, $Q^{AB/F}$	Unclear
DWT [189]	2015	PET-MRI	DWT, FCM, IHS Transform	**AANLIB**	MSE, PSNR, SD, AG	Unclear
MST-SR[61]	2015	CT-MRI, Gd DTPA MRI-MRI T2	MST, Sparse Representation, OMP	Private	SD, EN, QG, QP, QW	Unclear
Surface regis[190]	2015	CT-US	EM System, ICP Algorithm	Private	RMS	Unclear
QC hybrid [156]	2015	CT-MRI	DWT, PCA	Private	LM, E, MI, visual evaluation	Unclear
SOMP [191]	2016	MRI-CT, MRI-MRA, MRI-PET	K-SVD, SOMP, Sparse Representation	**AANLIB**, **Imagefusion.org**	EN, EI, $Q^{AB/F}$, MI, VIF	Unclear
Wavelet channel fusion [160]	2017	MRI T1-CT, MRI DWI-CT	DWT, Mallat Algorithm, Max Selection	Human Connectome Project	IE, MI, MG, SF, visual evaluation	Unclear
PCA-DWT [161]	2017	CT-MRI, MRI-SPECT	PCA, DWT, PCNN	**AANLIB**	STD, E, PSNR, MI, $Q^{AB/F}$	Unclear
NSCT-PCNN[192]	2017	CT-PET	NSCT, PCNN, NSML	CT	AG, EN, IQI, $Q^{AB/F}$	Unclear
Pyramid CNN[65]	2017	CT-MRI	Siamese CNN, image pyramids	SPECT	EN, FMI, Q_G, Q_E, VIFF	Unclear
MFDF-NSST[193]	2018	CT-MRI	NSST, MFDF, SML, Max Selection	**AANLIB**, **Imagefusion.org**	MI, EN, Q_E	Unclear
CNN Type-II (feature)[32]	2018	PET-CT-MRI T1/T2	Separate 2D CNN encoders; shared classifier	**TCIA**	Pixel-wise Accuracy, visual evaluation	Benchmark
MANOVA [194]	2018	fMRI-sMRI-DTI	Multivariate analysis of variance	Private	Jaccard, Kappa, Wilk's lambda, ROC, AUC	Unclear (single-center cohort)
DCHWT-PCNN [195]	2018	MRI-PET, MRI-SPECT	DCHWT, PCNN	**AANLIB**	AG, STD, SSIM, $Q^{AB/F}$, MI, FSIM	Unclear
SKR-LDP [64]	2018	PET-MRI, SPECT-MRI	Sparse Representation, OMP, PCA, K-SVD	**AANLIB**	MI, $Q^{AB/F}$, VIF, Q_{CB}	Benchmark
Fuzzy-RBFNN [196]	2018	CT-MRI, MRI-SPECT	Fuzzy-RBFNN, EBPA-GSA, PCNN	Private	EN, STD, MI, $Q^{AB/F}$, MSE	Unclear
Tetrolet SUnSAL [63]	2018	CT-MRI, PET-MRI, SPECT-MRI	SUnSAL, Tetrolet Transform, Cycle-spinning	**AANLIB**, **Imagefusion.org**	MI, LSSIM, $Q^{AB/F}$, FSIM, FSIMC	Unclear
MDWT [197]	2018	CT-PET, SPECT, MRI, MRA, MRT	MDWT, PCA, Max Selection	Private	CE, H, SD, SF, FQI, RO, AVGER	Unclear
FCN [198]	2019	PET-CT	FCN, V-Net	Private	DSC, CE, VE	Unclear
NSCT [199]	2019	CT-MRI, MRI-PET, SPECT-MRI	NSCT, YUV Transform	**AANLIB**	TE, IE, $Q^{AB/F}$, Q_P, MI, Q_Y, Q_{CB}, VIF	Unclear
MCFO [200]	2019	MRI-PET, CT-PET, MRI-CT	DWT, MCFO, PCA	MedPix Dataset	AVG, L, C, STD, E, PSNR, $Q^{AB/F}$	Unclear
NSST-SPCNN[201]	2020	CT-MRI, CT, MRI	NSST, SPCNN , RE, EOG	Private	SD, EN, MI, SF, Q_G, Q_{CB}	Unclear
FCM-PCA CNN[202]	2019	CT-MRI, PET-MRI, SPECT-MRI	CNN, FCM, PCA	AANLIIB	EN, MI, STD, UIQI, FMI, SF	Unclear
Multi-level fusion [203]	2020	PET-CT	Image-, matrix-and feature-levels fusion	**TCIA**	C-index, Spearman, log-rank p, Kaplan-Meier	Unclear (multicenter design)
IVD-Net [204]	2019	MRI multi-sequence	U-Net, HyperDenseNet, Inception	MICCAI 2018 challenge	DSC	Unclear
Patch CNN[205]	2019	PET-CT-MRI T1, PET-CT-MRI T2	multi-stage CNN fusion	STS	DSC, Acc	Unclear

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Table S2 continued

Paper	Year	Modalities	Method	Datasets	Metrics	Validation evidence
ZL CNN[206]	2019	MRI-CT/SPECT/PET, MRI T1-MRI T2-CT-PET	CNN, VGG-16, NSCT, PCNN	**AANLIB**	EN, MI, SSIM, $Q_{MRI-PET}$	Unclear
SWT-FL[207]	2019	CT-MRI, MRI T1-T2, MRI T2-SPET	SWT, FLICM	Private	MI, $Q^{AB/F}$, BSSIM, SSIM	Unclear
Siamese CNN [170]	2020	MR-CT, MR-T, MR-PET, MR-SPECT	CNN, contrast pyramid	**AANLIB**	$Q^{AB/F}$, MI, VIF	Unclear
U2Fusion[208]	2022	PET-MRI	Unsupervised loss, VGG, EWC	**AANLIB**	CC, SSIM, PSNR, SCD, EI, VIF, MG, visual evaluation	Unclear
NSML-NSST[209]	2020	CT-MRI, MRI T1-T2, MRI T2-GAD, MRI-PET, MRI-SPECT	NSST, YIQ Transform, Mamdani	**Imagefusion.org**, **AANLIB**	EN, FMI, $Q^{AB/F}$, SD, MSSIM	Benchmark
MM ASeDiL[210]	2020	CT-MRI, MRI T1-MRA, MRI Gd-T2, MRI T1-T2	NSCT, ASeDiL, Gabor Filter	**AANLIB**	Piella–Heijmans, SF, UIQI, MI	Unclear
DWT0-BCSO[211]	2020	MRI-CT	DWT, BCSO	**AANLIB**	EN, SSIM, FF	Benchmark
Auto CNN[113]	2020	MRI-CT	SIFT, DWT, Curvelet Transform, interpolation, CNN	Private	Acc, visual evaluation	Unclear
NMHE-SSGSM[62]	2020	CT-MRI	CSR, Average Rule, Max Selection	**Imagefusion.org**, **AANLIB**	MI, $Q^{AB/F}$, FMI, EN, VIF	Unclear
IFCNN[212]	2020	CT-MRI	IFCNN, CNN, ResNet101	Private	VIFF, ISSIM, NMI, SF, AG	Unclear
LRD [213]	2020	PET-MRI, SPECT-MRI	LRD, OD, NOD, IRS, LEM, LDM	**AANLIB**	STD, MI, UQI, TMQI	Unclear
3D MM-NAS [94]	2020	PET-CT	MM-NAS, CNN	**TCIA**	Acc, Prec, F1, AUC, ROC	Benchmark
NSST+BM-PCNN [214]	2025	MRI-PET, SPECT	NSST, BM-PCNN, MSMG	**AANLIB**	EN, SD, NMI, VIF, Q_F	Unclear
MMFNet[215]	2020	MRI T1-T2-T1ce	MMFNet, 3D-CBAM, RFBlock	Private	DSC, ASD, HD	Unclear (single-center cohort)
Res-DenseNet [216]	2020	MRI T1-T1ce-T2-FLAIR	DenseNet, ResNet	**BraTS2015, BraTS2018**	DSC	Benchmark
MGMDcGAN[101]	2020	MRI-PET, MRI-SPECT, CT-SPECT	cGAN, mask	**AANLIB**	EN, SF, MG, PSNR, SSIM	Unclear
Bilevel fusion[82]	2021	CT-MRI, MRI-SPECT	CNN, Transformer bilevel optimization	**AANLIB**	VIF, AG, EN, SF, SCD, $Q^{AB/F}$	Unclear
DDcGAN[81]	2020	MRI-PET	DDcGAN, dual-discriminator	**AANLIB**	EN, MG, SF, SD, PSNR, CC, SSIM, VIF	Benchmark
DBED [217]	2021	CT-MRT, MRGAD-MRT, CT-MRPD	GAN, DBED	**AANLIB**	MI, SSIM, NCIE, Chen-Blum, visual evaluation	Benchmark
NSCT-Xception[218]	2021	MRI, CT	NSCT, Xception image reconstruction	**Imagefusion.org**, **AANLIB**	API, EN, AG, MI, FS, CC, SF, RMSE, PSNR	Benchmark
EMFusion[2]	2021	CT, MRI, PET, SPECT	FusionNet, CNN	**AANLIB**	SSIM, PSNR, CC, VIF	Benchmark
EOA Gabor energy[219]	2021	MRI-PET	DSWT, FR-MGE, EOA	**AANLIB**	ALI, CI, S, $Q^{AB/F}$, VIFF, FMI	Unclear
BMGAN[56]	2021	MRI-PET	3D Dense-UNet, GAN	ADNI, OASIS-3, **TCIA**	MAE, PSNR, MS-SSIM, FID	Unclear
3D CNN[220]	2021	MRI-fMRI-DTI	3D-CNN	COBRE	ACC, SEN, SPE, AUC	Unclear
DTNP-NSCT[221]	2021	CT-MRI, MRI T1-T2, MRI-PET/SPECT	NSCT, INSML, WLE, DTNP	**AANLIB**	Petrovic, MI, FMI, SCD, MS-SSIM, SD	Unclear
NSCT-CNN[222]	2021	CT-MRI, MRI multi-sequence	NSCT, CNN, PHF-CNN, FCN-8s	**AANLIB**	MI, SSIM, VIF, $Q^{AB/F}$	Unclear
CBF-ANFIS [218]	2021	CT-MRI, MRI GAD-T2, CT-MR GAD, MR GAD-PET-FDG	CBF, ANFIS, DWT, Average Rule	**AANLIB**	API, AG, EN, MIF, FS, CC, SF, PSNR, $Q^{AB/F}$	Benchmark
NSST-Co-LE[60]	2021	CT-MRI, MRI-PET	NSST, LE, CoF, MSS, SML, Max Selection	**AANLIB**	MI, SD, $Q^{AB/F}$, SF, EN	Benchmark
DCT [53]	2021	CT-MRI	DCT, DWT	Private	SNR, PSNR, MSE, AG, SD, EN	Unclear

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Table S2 continued

Paper	Year	Modalities	Method	Datasets	Metrics	Validation evidence
Aggregated NAS[95]	2021	MRI-PET	NAS, layered fusion	**AANLIB**	SD, EN, VIF, CC, SCD	Unclear
DPCN [223]	2021	SPECT-MRI	CNN, SFFEB, FSFEB	**AANLIB**	MI, VIF, FMI, Q_P, Q_Y, time	Unclear
MSRPAN [104]	2021	MRI-CT, MRI-PET, SPECT	CNN, MSRPAN, FER	**AANLIB**	Q_{AC}, SSIM, MI, PSNR, Q_{AC}, TMQI, FSIM, time	Unclear
SDNet[224]	2021	PET-MRI	SDNet, adaptive decision	**AANLIB**	EN, $Q^{AB/F}$, FMI, PSNR, MG, time, visual evaluation	Unclear
PPT fusion[225]	2021	MRI-PET	Patch/Pyramid Transformer, MLP	**AANLIB**	EN, SSIM, MS-SSIM, MI, FMI, SCD, CC, SD, VIF, visual evaluation	Unclear
IGNFusion [226]	2022	CT-MR T1, PET-MRI T2	VGG-16, MSCAFM/MSCAFM, IGM, SSW	**AANLIB**	NMI, $Q^{AB/F}$, Q_Y, UQI	Unclear
DB-NSCT [227]	2022	CT-MRI, MRI-PET	NSCT, Davies-Bouldin	**AANLIB**	MI, SSIM, $Q^{AB/F}$	Unclear
FDO [228]	2023	MRI-PET	SVD, WMC, FDO, DPGF	**AANLIB**	AVG, MI, $Q^{AB/F}$, VIF, SSIM, PSNR	Unclear
COVID-19 CNN[229]	2022	CT-X ray	AlexNet, Vgg16, GoogLeNet	Private	ACC, SEN, SPEC, MCC, AUC	Unclear
CRFNet [111]	2024	Acetic-acid–iodine colposcopy	Proposal shifting; corresponding channel attention	Private	Average precision	Unclear
ST-cGAN [57]	2022	MRI T1-T2	cGAN	IXI	PSNR, SSIM, FSIM, MMD	Unclear
MUM-Net [230]	2022	MRI-US	MUM-Net, Disentangling modality features	Private	AUC, ACC, SENS, SPEC, PREC, Grad-CAM	Unclear
ADSR [231]	2022	MRI T2+FDG-PET, MRI T2+SPECT-TI/TC	Sparse Representation, PCA, K-SVD	**AANLIB**	EN, AG, MI, Q_W, $Q^{AB/F}$	Unclear
IoMT-NSST [232]	2022	CT-MRI, MRI-SPECT	NSST, MLE	**AANLIB**	MI, SD, $Q^{AB/F}$, SF, Mean, AG, CC, time	Benchmark
MATR[117]	2022	CT-MRI, MRI-SPECT/PET	Multiscale adaptive Transformer, composed loss	**AANLIB**	MI, TE, NICE, G, P, CV, LMI, VIF, MS-SSIM	Unclear
DualMMP-GAN[46]	2022	MRI T1-T2/FLAIR	DualMMP-GAN, GMMS	**BraTS 2018**	PSNR, SSIM, RMSE, DSC, Sensitivity, Spesificity, HD95	Benchmark
MMRACR-Net [233]	2022	MRI DWI-ADC	MMRAM, CCM	Private	Acc, AUC, SEN, SPC	Unclear
TAUNet[234]	2022	MRI bSSFP-LGE	CNN, Transformer, OCR	MyoPS 2020	DSC, HD95	Benchmark
SwinFusion[34]	2022	PET-MRI, CT-MRI	CNN, Swin Transformer, ACFM	**AANLIB**	FMI, $Q^{AB/F}$, SSIM, PSNR	Benchmark
DILRAN[235]	2022	CT-MRI	CNN, DILRAN, Softmax	**AANLIB**	PSNR, SSIM, FSIM, MI, FMI, EN	Benchmark
IFT [236]	2022	MRI-PET	Spatio-Transformer, RFN-Nest	**AANLIB**	EN, SCD, MI, MS-SSIM, SD, CC, MG	Unclear
MRSCFusion[118]	2023	CT-MRI, PET-MRI, SPECT-MRI	Swin Transformer, CNN, RSCF, AWB	**AANLIB**	SCD, $Q^{AB/F}$, FMI, SF, VIF	Unclear
DFENet[237]	2023	CT-MRI, PET-MRI, SPECT-MRI	CNN, Transformer, DFENet, GSFAM, LEGM	**AANLIB**	MI, FMI, NICE, SSIM, $Q^{AB/F}$, CV, Q_{CB}	Benchmark
Hybrid Transformer[238]	2023	PET-MR, SPECT-MR	CNN, Transformer, self-adaptive weight fusion rule	**AANLIB**	MI, Q_P, Q_{NICE}, SSIM	Benchmark
F²Net[239]	2023	MRI multi-sequence	Transformer, PVTv2, F²Net, CFM, MCM	**BraTS 2019, BraTS 2020**, ISLES 2015 SISS	DSC, IoU, HD95, Sen	Unclear
Epsilon GFS [240]	2023	CT-MRI	Shearlet Transform, Guided Filter, Max Selection, Average Rule	Public MRI-CT dataset	$Q^{AB/F}$, L^F_{AB}, N^F_{AB}, MAD, SD	Unclear
U-Patch GAN[86]	2023	SPECT TI-MRI T2, SPECT TC-MRI T2, CT-MRI T2, PET FDG-MRI T1	PatchGAN, spectral norm	**AANLIB**	Q_G, Q_Q, FMI, SSIM, RMSE, MI	Unclear
ELF[241]	2023	2D Fundus-3D OCT	CNN, ELF	GAMMA	Acc, Cohen's Kappa	Benchmark
AMTNet [242]	2023	MRI T2-ADC, MRI T1-FLAIR	Transformer, CNN, AMTNet, R-MSA, AITF, 3D-CDS	**BraTS 2021**	DSC, HD95, Jaccard, RVD	Benchmark

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Table S2 continued

Paper	Year	Modalities	Method	Datasets	Metrics	Validation evidence
DDFM[71]	2023	MRI-CT, MRI-PET, SPECT-MRI	DDPM, hierarchical Bayesian inference	**AANLIB**	EN, SD, MI, VIF, $Q^{AB/F}$, SSIM	Benchmark
MSU-Net[243]	2023	MRI T1-T2-T1ce	U-Net, MSU-Net, SCIEM, MSFEM	**AANLIB**, private	DSC, PREC, Recall, IOU	Unclear
CDDFuse[9]	2023	MRI-CT, MRI-PET, MRI-SPECT	CNN, Restormer, INN, CDDFuse	**AANLIB**	EN, SD, SF, MI, SCD, VIF, $Q^{AB/F}$, SSIM	Benchmark
IE-CFRN[244]	2023	CT-MRI T1/T2, MRI T1-GAD, MRI T1-PD, MRI-PET/SPECT TC /SPECT TI	CNN, GCIE, HFEN, LWEN	**AANLIB**	$Q^{AB/F}$, VIFP, IFC, NMI	Unclear
CAE+MCIEN[10]	2023	CT-MRI T1/T2, MRI T1-T2, MRI T2-GAD, MRI T2-PD, MRI T2-PET/SPECT TC/SPECT TI	CNN, Vision Transformer, CAE, MCIEN	**AANLIB**	NMI, MI, FSIM, PC, GM, $Q^{AB/F}$, UIQI	Unclear
3D THERMO-SCAN[245]	2023	CT-Infrared	CNN, 3D THERMO-SCAN	Private	Registration error, visual evaluation	Unclear
TUFusion[39]	2024	Medical and other image domains; exact medical pairing unverified	CNN–Transformer encoder; composite attention	Medical subset not verified	PSNR, RMSE, SSIM	Unclear
SSTFusion [246]	2023	CT-MRI	LeWin Transformer, CNN, RSR pretext	**AANLIB**	MI, PSNR, $Q^{AB/F}$, SSIM, AG, EI, SF	Unclear
ALMF-Net[96]	2024	MRI-PET/CT/SPECT	NAS, DARTS, mask	**AANLIB**, **BraTS 2021**	MI, CC, $Q^{AB/F}$, VIF, P, FLOP, L	Unclear
MUFusion[36]	2023	PET-MRI	VGG-19, memory unit	**AANLIB**	VIF, EN, MI, EI, $Q^{AB/F}$, MSSIM, DSC	Unclear
M3AE [12]	2023	MRI T1-T2-T1ce-FLAIR	M³AE, self distillation, model inversion	**BraTS 2018/2020**	DSC, HD95	Benchmark
FDGNet[247]	2023	CT-MRI, PET-MRI, SPECT-MRI	CNN, FDGNet	**AANLIB**	NMI, QY, $Q^{AB/F}$, VIF	Benchmark
MURF[114]	2023	PET-MRI, CT-MRI	Mutually reinforcing registration-fusion, Shared Information Extraction via contrastive learning, Multi-scale Coarse Registration	**AANLIB**	RMSE, MAE, MEE, NCC, MG, EI, VIF, SSIM, MI	Unclear
M4FNet[98]	2023	CT-MRI, PET-MRI, SPECT-MRI	CNN, M4FNet	**AANLIB**, **BraTS 2019**	EI, SD, $Q^{AB/F}$, MI, MSSIM, PC, VIF, UIQI	Unclear
FATFusion[67]	2024	SPECT-MRI, PET-MRI	Transformer, FGTM, AGTM	**AANLIB**	NMI, QY, $Q^{AB/F}$, VIF, Chen-Blum	Unclear
MCF-Net[91]	2024	MRI multi-sequence	ViT, mask align, sobel	**BraTS 2020**	PSNR, SSIM, NMSE	Unclear
MACTFusion[105]	2025	CT-MRI, PET-MRI, SPECT-MRI	ViT, CNN multi-axis attention	**AANLIB**	SCD, SD, MS-SSIM, $Q^{AB/F}$, VIF, Q_w	Unclear
F-DARTS[248]	2023	CT-MRI, MRI-PET, MRI-SPECT, multi-sequence MRI	Unsupervised differentiable architecture search, HVS-aligned kernel learning, unsupervised loss, channel attention in search space	**AANLIB**, BrainWeb, RIRE	MI, FMI_w, $Q^{AB/F}$, SCD, VIF, SD, Q_{CB}	Unclear
ALYOLOv5 [7]	2024	PET-CT	YOLOv5, CFF , AAF	Private	Precision, recall, mAP, F1	Unclear (single-center cohort)
MGCNet[249]	2024	Multi-sequence MRI	Guided local feature fusion; cross-modal complementary external attention	Private	ACC, AUC, C-index, HR	Unclear
EMMA [250]	2024	MRI-CT, MRI-PET, SPECT-MRI	U-Net, Restormer, EMMA	**AANLIB**	EN, SD, SF, AG, SCD, VIF	Benchmark
SRMNet[251]	2024	MRI multi-sequence	Missing-modality segmentation, Local-global feature enhancement, Deformation-Adaptive Perception fusion	**BraTS 2018**, **BraTS 2020**	DSC, HD95	Benchmark

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Table S2 continued

Paper	Year	Modalities	Method	Datasets	Metrics	Validation evidence
CoCoNet[140]	2024	MRI-PET, MRI-SPECT	VGG, contrastive learning, CoCoNet	**AANLIB**	EN, AG, SF, SD, SCD, VIF	Benchmark
ASFE-Fusion [66]	2024	CT-MRI, PET-MRI, SPECT-MRI	CNN, CASF, AFF, SFEF, FFT	**AANLIB**	EN, PSNR, MI, CC, FMI, Prec, Recall, F1, Acc	Unclear
CA-DDPM [73]	2024	MRI-CT	DDPM, CNN, Transformer	**AANLIB**	EN, MI, SCD, SSIM, VIF, $Q^{AB/F}$	Benchmark
Diff-IF[90]	2024	MRI, CT, PET, SPECT	DDPM, prior knowledge, target search	**AANLIB**	MI, SCD, VIF, $Q^{AB/F}$, SSIM	Unclear
TIEF [252]	2024	MRI multi-sequence, SPECT-MRI, CT-MRI	CNN, SDCL, TIEF	**BraTS 2019**, **AANLIB**	$Q^{AB/F}$, EN, AG, SF, CI, MI, PSNR, SSIM, F	Unclear
Mambadfuse[70]	2024	MRI-PET, MRI-SPECT	Mamba, CNN	**AANLIB**	EN, SD, SF, MI, SCD, VIF, $Q^{AB/F}$, SSIM, mAP, Grand-CAM, visual evaluation	Benchmark
Fuse4Seg[108]	2024	MRI T1-FLAIR-T2-T1ce, MRI-CT/PET/SPECT	SwinUNet, dual-level optimization	**BraTS 2021**, **AANLIB**	EN, SD, SF, MI, SCD, VIF, $Q^{AB/F}$, DSC, mIoU, HD95, visual evaluation, attention heatmap	Benchmark
MHFN [107]	2024	MRI T1W–T2W-STIR	Hierarchical feature fusion with AGFM	Private	DSC, IoU, HD	Unclear (single-center cohort)
CDE-Net[84]	2024	MRI T1-T2-FLAIR-T1ce	CDE-Net, DST, CDFL, RW-PAM	**BraTS 2021**	DSC, HD95, visual evaluation	Unclear
TIMFusion[253]	2024	MRI-CT/PET/SPECT	Task guidance constraints, IAS, PMI	**AANLIB**	MI, FMI, VIF, $Q^{AB/F}$, EN, SCD, parameter, time	Benchmark
LRFNet[37]	2024	MRI-PET, MRI-SPECT	LRU, FEF, DGSSIM	**AANLIB**	EN, SD, MI, AG, CC, SCD, $Q^{AB/F}$, SF	Unclear
CMINet[254]	2025	CT-MRI, MRI-PET, MRI-SPECT	Transformer, CNN, GDA/GADN	**AANLIB**	MI, SF, VIF, $Q^{AB/F}$	Unclear
PDMINet[255]	2025	MRI multi-sequence	Prototype-driven + multiexpert integration	**BraTS 2018**, **BraTS 2019**, **BraTS 2020**	DSC, HD95, Sensitivity, Specificity	Unclear
GeSeNet[256]	2024	CT-MRI, MRI-PET, MRI-SPECT	Semantic-guided fusion, edge enhancement gradient filter, region mask for redundant computation pruning	**AANLIB**	SSIM, SD, MI, VIF, SCD, $Q^{AB/F}$	Unclear
DRMF[89]	2024	CT-MRI, MRI-PET, SPECT-MRI	Degradation-robust conditional diffusion, fusion + regularization + mask-guided loss	**AANLIB**	MI, SF, SD, VIF, SSIM, PSNR, MUSIQ, LIQE, mAP	Unclear
TFEFusion[257]	2025	PET, MRI	CNN, Swin Transformer, TFEFusion	**AANLIB**	G, SSIM, $Q^{AB/F}$, PSNR, SF, RMSE	Benchmark
LAGN Diffusion[124]	2025	MRI-CT, MRI-PET, MRI-SPECT	Diffusion, Transformer, CNN	**AANLIB**	EN, SD, SCD, VIF, PSNR, SSIM	Benchmark; Reader study (student ratings)
PFRFusion [5]	2025	CT-MRI, MRI T1-T2	PFAR, DBFF, co-training	**AANLIB**, Synth-Strip	MAE, NMI, MEE, MI, VIF, SSIM, IMF, SF, DSC, IoU, MPA	Unclear
EFCNet [92]	2025	MRI-SPECT, MRI-PET	EFCNet, SEEM, MSAM, E-Fusion, IRM	**AANLIB**	AG, SNR, MI, $Q^{AB/F}$, VIF, PSNR, EI, SF	Unclear
DTMFusion [97]	2025	CT-MRI, MRI T1-T2	Transformer, Mamba, DTSDR, MCF	**AANLIB**, Synth-Strip	PSNR, NCC, MSE, SD, MI, VIF, SCD	Unclear
EDPSN [125]	2025	MRI-CT, MRI-PET, SPECT-MRI	EDPSN, EEM, ED, DPS, VGG16	**AANLIB**, CHAOS	EN, SD, SCD, $Q^{AB/F}$, VIF, PSNR, SSIM	Benchmark; Unclear (reader protocol)
AAHN [258]	2025	PET-CT, MRI T1-FLAIR -DWI-ADC, CT-MRI	CNN, Transformer, GCN	Hecktor21, Prostate158, **BraTS 2019/2023/2024**, CHAOS	DSC, HD95	Unclear
MLFuse[259]	2025	CT-MRI, PET-MRI, SPECT-MRI	CNN, Transformer, sobel	**AANLIB**	VIF, SCD, $Q^{AB/F}$, SSIM, FMI	Unclear
DMCMFuse[68]	2025	CT-MRI, MRI T1-T2, MRI-PET/SPECT	CNN, Mamba, MCFM, HAED	**AANLIB**	SSIM, VIFF, $Q^{AB/F}$, SCD, FMI, Q_w	Unclear

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Table S2 continued

Paper	Year	Modalities	Method	Datasets	Metrics	Validation evidence
FAB-Transformer[260]	2025	PET-MRI, SPECT-MRI	CNN, Swin Transformer, DSM, INN	**AANLIB**	MI, SF, VIF, $Q^{AB/F}$, SSIM, mAP	Benchmark
AdvTCUnet [87]	2025	3D MRI T1-T1ce-FLAIR	Adversarial learning, Transformer, U-Net, CV energy function	**TCIA**	DSC, HD, visual evaluation	Benchmark
MTG-Fusion[31]	2025	CT-MRI, PET-MRI, SPECT-MRI	Auxiliary text guidance, image-feature fusion	**AANLIB**	EN, SD, SF, SCD, VIF, Q	Unclear
MAINet[261]	2025	MRI-CT, MRI-PET, MRI-SPECT	CMI, DGL, BUGF	**AANLIB**	MI, Q_{NCIE}, SD, VIF, $Q^{AB/F}$, SSIM, Q_{CB}, visual evaluation	Unclear
Uncertainty-seg[83]	2025	MRI FLAIR-T2	MC-Dropout, uncertainty weighting	BraTS subset	DSC, IoU, SE, PPV, uncertainty	Unclear
BSAFusion[11]	2025	Misaligned CT-MRI, PET-MRI	MDF-FR, BSFA, MMFF	**AANLIB**	$Q^{AB/F}$, Q_{CV}, SSIM, VIF, Q_S, visual evaluation	Benchmark
UniFuse[38]	2025	MRI T1-T2/CT, CT-PET	Spatial Mamba, LoRA, DAPL, OUFR	**BraTS 2020**, SynthRAD2023, FDG-PET/CT	MSE, CC, PSNR, VIF, SSIM, GFLOP, visual evaluation	Unclear
ReFusion[262]	2025	CT-MRI, MRI-PET	Meta-learning, learnable fusion loss, inner-outer-alternate training strategy	**AANLIB**	EN, SD, SF, AG, SCD, SSIM, VIF, Q_{CB}, Q_{NCIE}	Benchmark
DM-FNet[72]	2025	CT-MRI, MRI-PET, MRI-SPECT	Two-stage diffusion hybrid loss (intensity + SSIM + gradient)	**AANLIB**	SF, SD, AG, Q_W, SCD, VIFF, $Q^{AB/F}$, MSSSIM, FMI-WT	Unclear
C2RF[263]	2025	PET-MRI	Mutually promoting registration-fusion, commonality mining for feature disentanglement, multi-level registration, contrastive learning with hard negative sample mining	**AANLIB**	MI, CC, SSIM, MSE, MEE, MAE, EN, MI, SF, VIF, $Q^{AB/F}$, SSIM	Unclear
HCFMaNet[99]	2026	CT-MRI, PET-MRI, SPECT-MRI, MRI multi-sequence	Local-aware Mamba, Holistic Cross-modal Fusion Mamba, end-to-end unsupervised fusion	**AANLIB**, BrainWeb, **SynthRAD**	Q_{NCIE}, MI, Q_G, Q_{CV}, LMI, MS-SSIM, VIF, NCC, SSIM	Unclear
FlexiD-Fuse[264]	2026	CT-MRI, MRI-PET, MRI-SPECT, MR-T2/MR-Gad/PET, CT/MR-T2/SPECT	Diffusion model + hierarchical Bayesian modeling, Expectation-Maximization, ultra-lightweight Diffusion Fusion Mamba	**AANLIB**	SSIM, MI, PSNR, VIF, Q_S, Q_{CV}, FSIM, Q_{NCIE}	Benchmark

Table S3: Representative decision-level fusion studies. Validation evidence is defined in Section 6.4.

Paper	Year	Modalities	Method	Datasets	Metrics	Validation evidence
DST-cls[77]	1996	Double echo MRI	DS theory, mass function, composite hypothesis	Private	K, decision consistency, visual evaluation	Unclear
CNN Type-III (decision)[32]	2018	PET-CT-MRI T1/T2	CNN, Majority voting	**TCIA**	Pixel-wise Accuracy, visual evaluation	Benchmark
Patch CNN[205]	2019	PET-CT-MRI T1	Voting, CNN	STS	DSC, Pixel-wise Accuracy	Unclear
DCMA [112]	2020	MRI-PET	DCNN ensemble, Adaboost, dropout	**ADNI**	Acc, Sen, Spe, AUC	Benchmark
Evidence fusion[35]	2022	MRI T1-T1Gd-T2-FLAIR	DST, contextual discounting, MMEF module	**BraTS 2021**	DSC, HD, ECE, visual evaluation	Benchmark
ETSCL [79]	2024	CFP-OCT-Vascular	Supervised contrastive loss, DST-based fusion	GAMMA	Cohen's kappa, Accuracy	Benchmark
MHFN decision baseline [107]	2024	MRI T1W–T2W-STIR	Combines modality-specific segmentation outputs; baseline in the MHFN study	Private	DSC, IoU, HD	Unclear (single-center cohort)
Deep evidential fusion[3]	2025	PET-CT, MRI multi-sequence	DST, evidence mapping, contextual discounting	PET-CT lymphoma dataset, **BraTS 2021**	DSC, Brier score, NLL, ECE	Benchmark; Retrospective
MAMS[122]	2025	FFDM-DBT	cGAN, MAMS fusion	NFH, GDHCM	RMSE, PSNR, UQI, MS-SSIM, Acc, AUC, F1-Score	Retrospective (two centers)
Uncertainty-seg[83]	2025	MRI FLAIR-T2	MC-Dropout, uncertainty weighting	BraTS subset	DSC, IoU, SE, PPV, uncertainty	Unclear